	Polynomial Identities
	Find the values of the unknowns in each of the following.
1.	$A(x + 4)(x - 3) + B(x - 3) = 5x^2 + 4x - 57$ Ans: $[A = 5, B =- 1]$
2.	$A(2x + 7) + B(7 - 2x) = 7$ Ans: $\left[A = \frac{1}{2}, B = \frac{1}{2}\right]$
3.	$- 6x^2 - 17x + 28 = A(x + 5)(2 - x) + B(2 - x) + C$ Ans:$[A = 6, B =- 1, C =- 30]$
4.	$2x^2 + C = A(x - 1)^2 + B(x - 1) - 9$ Ans:$[A = 4, B = 4, C =- 11]$
5.	$(x + 2)\left(Ax^2 + Bx + C\right) = 4x^3 + 7x^2 - 5x - 6$ Ans:$[A = 4, B =- 1, C =- 3]$
6.	$3x^3 + 25x^2 - 21x + C = \left(Ax^2 + Bx - 1\right)(3x - 2)$ Ans:$[A = 1, B = 9, C = 2]$
7.	$(Ax + 1)(x + B)(1 - 3x) + C = 12 + 5x - 47x^2 - 12x^3$ Ans:$[A = 4, B = 4, C = 8]$
8.	$5x^4 + 7x^3 - 125x^2 + 74x = \left(Ax^3 - 3x + 1\right)(x + B)(x + 6) + C$ Ans:$[A = 5, B =- 4, C = 24]$